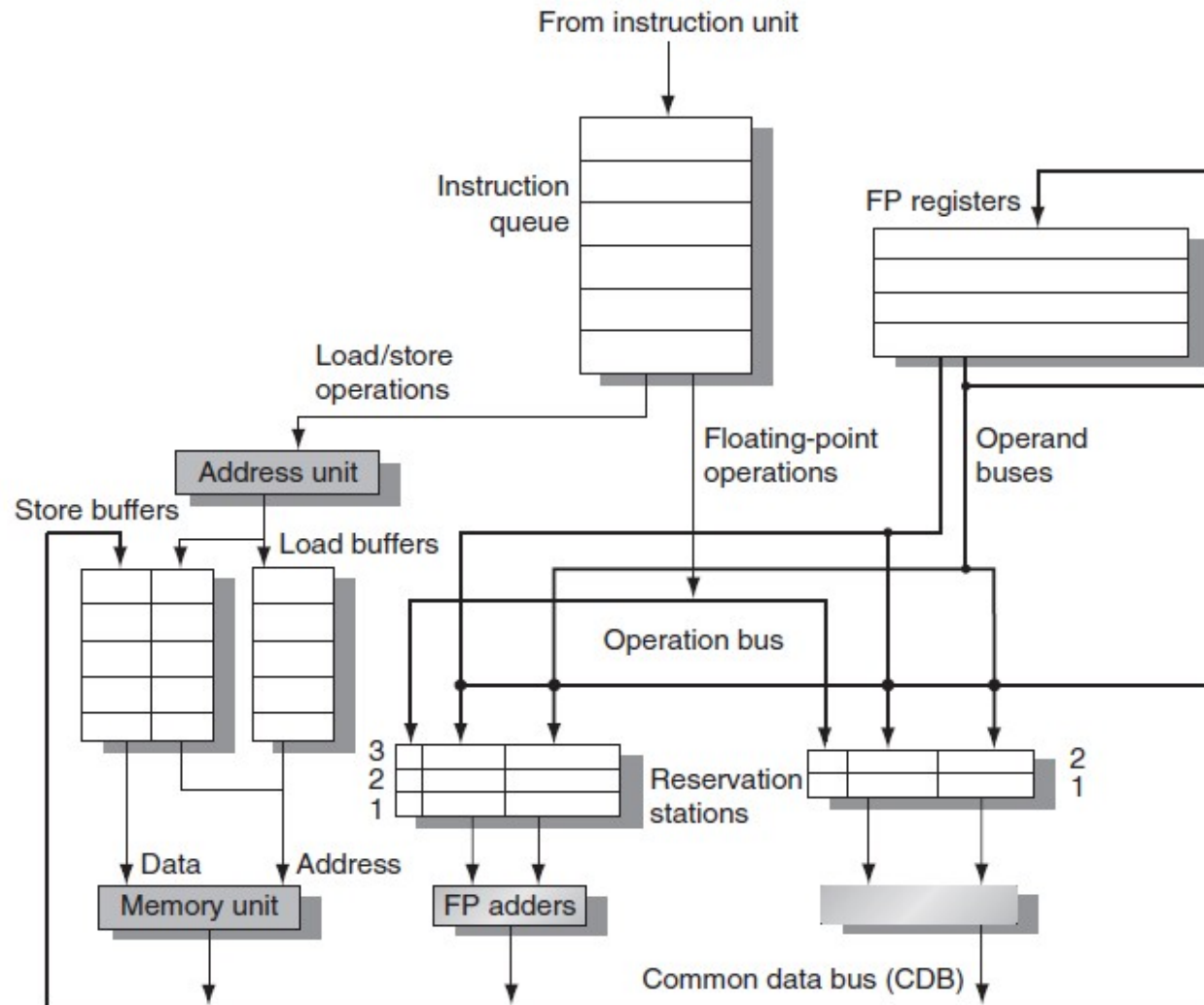


Algoritmo de Tomasulo - Ponto flutuante no MIPS



HENNESSY, John L.
Arquitetura de computadores uma abordagem quantitativa

9/15/22

Arquitetura de Computadores III - Engenharia e Ciência da Computação - PUC Minas

102

Instruction		Instruction status		
		Issue	Execute	Write result
L.D	F6,32(R2)	✓	✓	✓
L.D	F2,44(R3)	✓	✓	
MUL.D	F0,F2,F4	✓		
SUB.D	F8,F2,F6	✓		
DIV.D	F10,F0,F6	✓		
ADD.D	F6,F8,F2	✓		

Reservation stations							
Name	Busy	Op	Vj	Vk	Qj	Qk	A
Load1	No						
Load2	Yes	Load					44 + Regs[R3]
Add1	Yes	SUB		Mem[32 + Regs[R2]]	Load2		
Add2	Yes	ADD			Add1	Load2	
Add3	No						
Mult1	Yes	MUL		Regs[F4]	Load2		
Mult2	Yes	DIV		Mem[32 + Regs[R2]]	Mult1		

Register status									
Field	F0	F2	F4	F6	F8	F10	F12	...	F30
Qi	Mult1	Load2		Add2	Add1	Mult2			

Instruction		Instruction status							
		Issue	Execute	Write result					
L.D	F6,32(R2)	✓	✓	✓					
L.D	F2,44(R3)	✓	✓						
MUL.D	F0,F2,F4	✓							
SUB.D	F8,F2,F6	✓							
DIV.D		✓							
ADD.I		✓							
		Reservation stations							
Name	Busy	Op	Vj	Vk	Qj	Qk	A		
Load1	No								
Load2	Yes	Load					44 + Regs[R3]		
Add1	Yes	SUB		Mem[32 + Regs[R2]]	Load2				
Add2	Yes	ADD			Add1	Load2			
Add3	No								
Mult1	Yes	MUL		Regs[F4]	Load2				
Mult2	Yes	DIV		Mem[32 + Regs[R2]]	Mult1				
		Register status							
Field	F0	F2	F4	F6	F8	F10	F12	...	F30
Qi	Mult1	Load2		Add2	Add1	Mult2			

Estação de reserva está ocupada?

Instruction		Instruction status		
		Issue	Execute	Write result
L.D	F6,32(R2)	✓	✓	✓
L.D	F2,44(R3)	✓	✓	
MUL.D	F0,F2,F4	✓		
SUB.D	F8,F2,F6	✓		
DIV.D		✓		
ADD.I		✓		

Operação que será realizada!

Reservation stations							
Name	Busy	Op	Vj	Vk	Qj	Qk	A
Load1	No						
Load2	Yes	Load					44 + Regs[R3]
Add1	Yes	SUB		Mem[32 + Regs[R2]]	Load2		
Add2	Yes	ADD			Add1	Load2	
Add3	No						
Mult1	Yes	MUL		Regs[F4]	Load2		
Mult2	Yes	DIV		Mem[32 + Regs[R2]]	Mult1		

Register status									
Field	F0	F2	F4	F6	F8	F10	F12	...	F30
Qi	Mult1	Load2		Add2	Add1	Mult2			

Instruction		Instruction status		
		Issue	Execute	Write result
L.D	F6,32(R2)	✓	✓	✓
L.D	F2,44(R3)	✓	✓	
MUL.D	F0,F2,F4	✓		
SUB.D	F8,F2,F6	✓		
DIV.D	F10,F0,F6	✓		
ADD.D	F6,F8,F2	✓		

Estações de reserva que produzirão o resultado!

Reservation stations							
Name	Busy	Op	Vj	Vk	Qj	Qk	A
Load1	No						
Load2	Yes	Load					44 + Regs[R3]
Add1	Yes	SUB		Mem[32 + Regs[R2]]	Load2		
Add2	Yes	ADD			Add1	Load2	
Add3	No						
Mult1	Yes	MUL		Regs[F4]	Load2		
Mult2	Yes	DIV		Mem[32 + Regs[R2]]	Mult1		

Register status									
Field	F0	F2	F4	F6	F8	F10	F12	...	F30
Qi	Mult1	Load2		Add2	Add1	Mult2			

Instruction		Instruction status		
		Issue	Execute	Write result
L.D	F6,32(R2)	✓	✓	✓
L.D	F2,44(R3)	✓	✓	
MUL.D	F0,F2,F4	✓		
SUB.D	F8,F2,F6	✓		
DIV.D	F10,F0,F6			
ADD.D	F6,F8,F2			

Valor dos operandos fonte!

Name	Busy	Op	Vj	Vk	Qj	Qk	A
Load1	No						
Load2	Yes	Load					44 + Regs[R3]
Add1	Yes	SUB		Mem[32 + Regs[R2]]	Load2		
Add2	Yes	ADD			Add1	Load2	
Add3	No						
Mult1	Yes	MUL		Regs[F4]	Load2		
Mult2	Yes	DIV		Mem[32 + Regs[R2]]	Mult1		

Register status									
Field	F0	F2	F4	F6	F8	F10	F12	...	F30
Qi	Mult1	Load2		Add2	Add1	Mult2			

Instruction		Instruction status		
		Issue	Execute	Write result
L.D	F6,32(R2)	✓	✓	✓
L.D	F2,44(R3)	✓	✓	
MUL.D	F0,F2,F4	✓		
SUB.D	F8,F2,F6	✓		
DIV.D	F10,F0,F6	✓		
ADD.D	F6,F8,F2	✓		

1º: Valor imediato
2º: Endereço calculado

Reservation stations							
Name	Busy	Op	Vj	Vk	Qj	Qk	A
Load1	No						
Load2	Yes	Load					44 + Regs[R3]
Add1	Yes	SUB		Mem[32 + Regs[R2]]	Load2		
Add2	Yes	ADD			Add1	Load2	
Add3	No						
Mult1	Yes	MUL		Regs[F4]	Load2		
Mult2	Yes	DIV		Mem[32 + Regs[R2]]	Mult1		

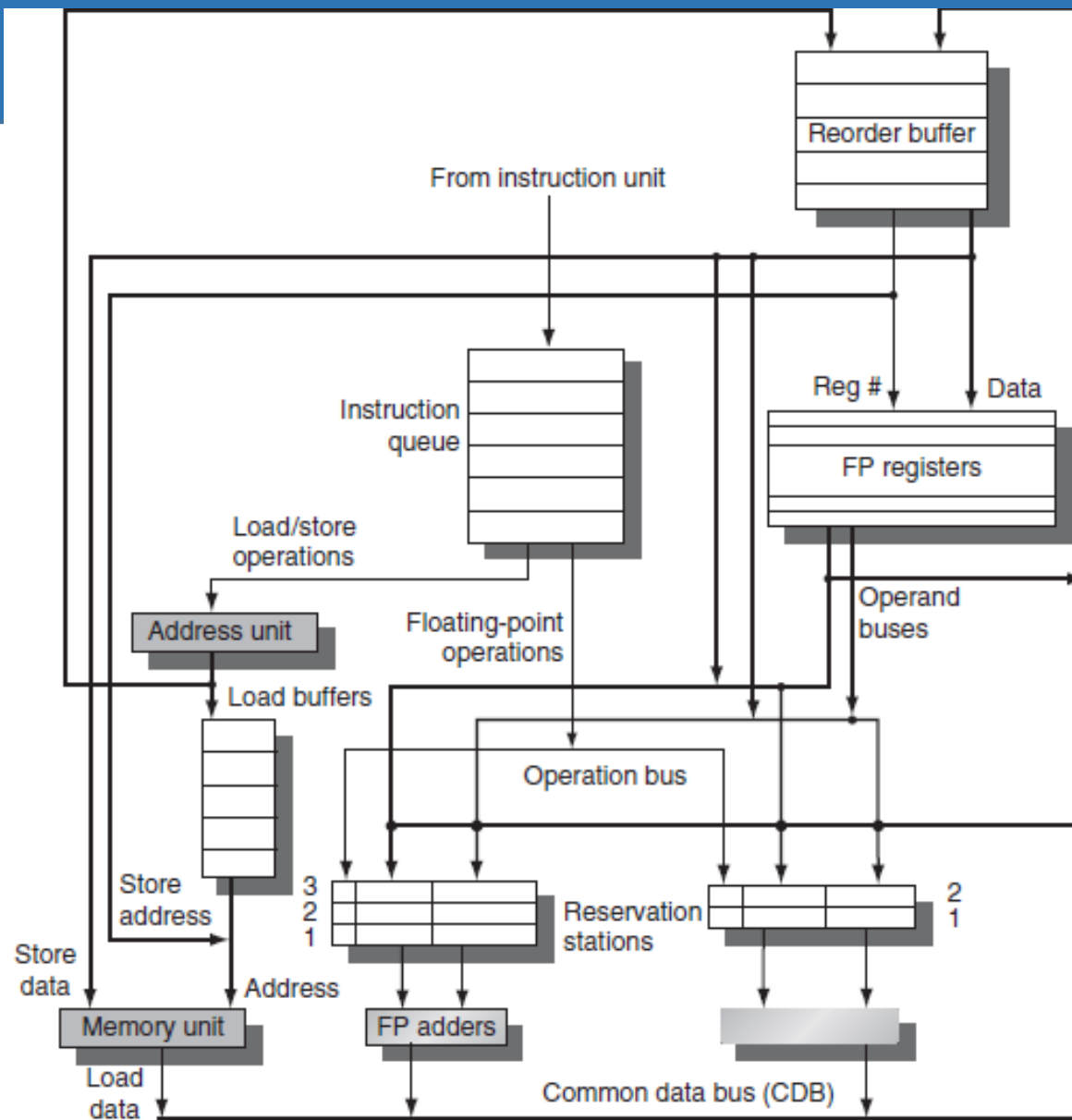
Register status									
Field	F0	F2	F4	F6	F8	F10	F12	...	F30
Qi	Mult1	Load2		Add2	Add1	Mult2			

Instruction		Instruction status		
		Issue	Execute	Write result
L.D	F6,32(R2)	✓	✓	✓
L.D	F2,44(R3)	✓	✓	
MUL.D	F0,F2,F4	✓		
SUB.D	F8,F2,F6	✓		
DIV.D	F10,F0,F6	✓		
ADD.D	F6,F8,F2	✓		

Reservation stations							
Name	Busy	Op	Vj	Vk	Qj	Qk	A
Load1	No						
Load2	Yes	Load					44 + Regs[R3]
Add1	Yes	SUB		Mem[32 + Regs[R2]]	Load2		
Add2	Yes	ADD			Add1	Load2	
				Regs[F4]	Load2		
				Mem[32 + Regs[R2]]	Mult1		

Nome da estação de reserva
que fornecerá o resultado ao
registrador

Register status									
Field	F0	F2	F4	F6	F8	F10	F12	...	F30
Qi	Mult1	Load2		Add2	Add1	Mult2			



Reorder buffer						
Entry	Busy	Instruction		State	Destination	Value
1	No	L.D	F6, 32 (R2)	Commit	F6	Mem[32 + Regs[R2]]
2	No	L.D	F2, 44 (R3)	Commit	F2	Mem[44 + Regs[R3]]
3	Yes	MUL.D	F0, F2, F4	Write result	F0	#2 × Regs[F4]
4	Yes	SUB.D	F8, F2, F6	Write result	F8	#2 − #1
5	Yes	DIV.D	F10, F0, F6	Execute	F10	
6	Yes	ADD.D	F6, F8, F2	Write result	F6	#4 + #2

Reservation stations								
Name	Busy	Op	Vj	Vk	Qj	Qk	Dest	A
Load1	No							
Load2	No							
Add1	No							
Add2	No							
Add3	No							
Mult1	No	MUL.D	Mem[44 + Regs[R3]]	Regs[F4]			#3	
Mult2	Yes	DIV.D		Mem[32 + Regs[R2]]	#3		#5	

FP register status										
Field	F0	F1	F2	F3	F4	F5	F6	F7	F8	F10
Reorder #	3						6		4	5
Busy	Yes	No	No	No	No	No	Yes	...	Yes	Yes